

1. Answer all the questions:

- (a) Find the radius and speed of the electron in the first Bohr orbit of the hydrogen atom. How will the radius and speed of electron change with the increase in atomic number of the atom?
- (b) In a Compton collision with an electron, a photon of violet light ($\lambda = 400 \text{ nm}$) is backward scattered through an angle 180° .
 - (i) How much energy is transferred to the electron in this collision?
 - (ii) Compare the result with the energy the electron would acquire in a photoelectric process with the same photon.
- (c) At what wavelength will the human body radiate the maximum radiation?
 - (i) Estimate the total power radiated by a person of medium build (assume an area given by a cylinder of 175 cm height and 13 cm radius).
 - (ii) Using your answer to (i), compare the energy radiated by a person in one day with the energy intake of a 2000 kcal diet.

2. Answer all the questions:

- (a) Calculate the impact parameter for scattering a 7.7 MeV α particle from gold at an angle of (i) 1° and (ii) 90° .
- (b) For aluminium ($Z = 13$) and gold ($Z = 79$) targets, what is the ratio of an alpha particle scattering at any angle for equal numbers of scattering nuclei per unit area?
- (c) Calculate the angular momentum in $\text{kg}\cdot\text{m}^2/\text{s}$ for the lowest electron orbit in the hydrogen atom. Compare the result with Planck's constant h . (b) Repeat for an electron in the $n = 2$ state of hydrogen.
- (d) A hydrogen atom in an excited state absorbs a photon of wavelength 410 nm. What were the initial and final states of the hydrogen atom?

3. Answer all the questions:

- (a) A beam of electrons is incident on a potential barrier 5 eV high, 0.5 nm wide. What energy should they have if half of them are to get through the barrier?
- (b) A particle of mass m is constrained to move in a 1D potential well with $V(x) = 0$ for $0 \leq x \leq L$ and $V(x)$ infinite elsewhere. The function is $\psi(x) = a \cos(n\pi x/L) + b \sin(n\pi x/L)$ where a & b are constants, and n is number. (i) Evaluate the constant b that ensures correct normalisation of the wave function $\psi(x)$. (ii) Sketch $\psi(x)$ and $\psi(x)^2$ for the lowest three values of n . (iii) An electron is confined to a 1D box of length 15 nm. How many energy levels lie between 3.5 and 8 KJ/mol $^{-1}$ ($m_e = 9.110 \times 10^{-31} \text{ kg}$).
- (c) From the Schrodinger equation for 1D harmonic Oscillator, (i) Find the wave function of the equation and also (ii) Find the energy associated with the obtained wavefunction.

4. Answer all the questions:

- (a) Assume that the electron in the hydrogen atom is constrained to move only in a circle of radius a in the xy plane.
 - (i) Show that the separated Schrödinger equation for ϕ becomes

$$\frac{1}{a^2} \frac{d^2 \psi}{d\phi^2} + \frac{2m}{\hbar^2} |E| \psi = 0$$
 where ϕ is the angle describing the position on the circle.
 - (ii) Explain why this is similar to the Bohr assumption.
 - (iii) Solve the above equation for ψ . Find the allowed energies and angular momenta. Compare your results with the Bohr theory.

- (b) Show that the radial wave function R_{21} for $n = 2$ and $l = 1$ satisfies equation given below

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2\mu}{\hbar^2} \left[E - V - \frac{\hbar^2 \ell(\ell+1)}{2\mu r^2} \right] R = 0$$

What energy results? Is this consistent with the Bohr model?

- (c) Show that the radial wave function R_{21} for $n = 2$ and $l = 1$ is normalized.